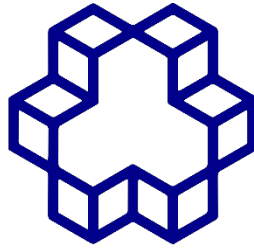


به نام خدا



دانشگاه صنعتی خواجه نصیرالدین طوسی

درس یادگیری ماشین

حل تمرینات سری اول (رگرسیون)

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دانشگاه صنعتی خواجه نصیرالدین طوسی – دانشکده مهندسی کامپیوتر

پاییز ۱۴۰۴

Question 1 (20 marks)

$n = 26$

The weight and systolic blood pressure of 26 randomly selected males in the age group 25 to 30 are shown in the following table. Assume that weight and blood pressure are jointly normally distributed. Find a regression line relating systolic blood pressure to weight.

من مراجعه مقدار فشار خون فرد را بر اساس وزن او تخمین بزنم و پیش بینی کنم. فرض کنید x مقدار وزن (weight) و y مقدار متناظر فشار خون فرد (Systolic blood pressure) باشد. چون متغیر Feature داریم (x) و $bias$ مزاحمت شده و توزیع داده ها نرمال (گادس) است می توانیم از فرمول های زیر برای حل رگرسیون خطی تک متغیره استفاده کنیم:

i	x	y		
Subject	Weight	Systolic BP	$x_i y_i$	x_i^2
1	165	130	21450	27225
2	167	133	22211	27889
3	180	150	27000	32400
4	155	128	19840	24025
5	212	151	32012	44944
6	175	146	25550	30625
7	190	150	28500	36100
8	210	140	29400	44100
9	200	148	29600	40000
10	149	125	18625	22201
11	158	133	21114	24964
12	169	135	22815	28561
13	170	150	25500	28900
14	172	153	26316	29584
15	159	128	20352	25281
16	168	132	22176	28224
17	174	149	25926	30276
18	183	158	28914	33489
19	215	150	32250	46225
20	195	163	31785	38025
21	180	156	28080	32400
22	143	124	17742	20449
23	240	170	40800	57600
24	235	165	38775	55225
25	192	160	30720	36864
26	187	159	29733	34969
Sum			697074 ①	110545 ②

$$\sum_{i=1}^n x_i y_i = 697074 \quad (1)$$

$$\sum_{i=1}^n x_i^2 = 110545 \quad (2)$$

$$w = \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2}$$

$$= \frac{697074}{110545} \approx 0.7919$$

$$\hat{y} = \text{out}(x) = wx$$

$$\hat{y} = 0.7919x$$

Question 2 (30 marks)

A study was performed on the wear of a bearing y and its relationship with x_1 oil viscosity and x_2 load. The following data were obtained.

(a) Fit a multivariate linear regression model to these data. (10 Marks)

(b) Use the model to predict wear when $x_1 = 25$ and $x_2 = 1000$. (5 Marks)

(c) Fit a multivariate linear regression model with an interaction term ($x_1 x_2$) to these data. (15 Marks)

y	x	
y	x_1	x_2
293	1.6	851
230	15.5	816
172	22.0	1058
91	43.0	1201
113	33.0	1357
125	40.0	1115

2-a Multivariate Regression

Least Squares

$$\omega = (X^T X)^{-1} X^T y$$

$$\hat{y} = \omega^T X$$

$$\omega = \begin{bmatrix} \omega_0 \\ \omega_1 \\ \omega_2 \end{bmatrix}$$

$$X = \begin{bmatrix} 1 \\ x_1 \\ x_2 \end{bmatrix}$$

bias x_0 مصروف به x $\rightarrow$ $X = \begin{bmatrix} x_0 & x_1 & x_2 \\ 1 & 1.6 & 851 \\ 1 & 15.5 & 816 \\ 1 & 22 & 1058 \\ 1 & 43 & 1201 \\ 1 & 33 & 1357 \\ 1 & 40 & 1115 \end{bmatrix}$

$X^T X = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1.6 & 15.5 & 22 & 43 & 33 & 40 \\ 851 & 816 & 1058 & 1201 & 1357 & 1115 \end{bmatrix}$ $X = \begin{bmatrix} 1 & 1.6 & 851 \\ 1 & 15.5 & 816 \\ 1 & 22 & 1058 \\ 1 & 43 & 1201 \\ 1 & 33 & 1357 \\ 1 & 40 & 1115 \end{bmatrix}$ $\therefore X^T X$ مصروف به X

$$A = X^T X = \begin{bmatrix} 7 & 120,1 & 4391 \\ 120,1 & 5248,11 & 1712,97 \\ 4391 & 1712,97 & 734894 \end{bmatrix}$$

② $(X^T X)^{-1}$ محاسبه

$$A^{-1} = (X^T X)^{-1} = \begin{bmatrix} 1,292 & 0,001 & -0,0099 \\ 0,001 & 0,002 & -0,0001 \\ -0,0099 & -0,0001 & 0,00001 \end{bmatrix}$$

③ $X^T y$ محاسبه

$$B = X^T y = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 14 & 12,2 & 22 & 23 & 33 & 20 \\ 151 & 114 & 1,15 & 12,1 & 1257 & 116 \end{bmatrix} \begin{bmatrix} 492 \\ 220 \\ 172 \\ 91 \\ 112 \\ 125 \end{bmatrix} = \begin{bmatrix} 1048 \\ 40209,1 \\ 1021004 \end{bmatrix}$$

④ محاسبه بردار $\vec{w}$ نهایی

$$\vec{w} = (X^T X)^{-1} X^T y = A^{-1} B = \begin{bmatrix} 1,292 & 0,001 & -0,0099 \\ 0,001 & 0,002 & -0,0001 \\ -0,0099 & -0,0001 & 0,00001 \end{bmatrix} \begin{bmatrix} 1048 \\ 40209,1 \\ 1021004 \end{bmatrix}$$

$$\Rightarrow \vec{w} = \begin{bmatrix} 313,1010 \\ -3,438 \\ -0,114 \end{bmatrix}$$

د) $\hat{y}$

$$\hat{y} = W^T X = \begin{bmatrix} w_0 \\ w_1 \\ w_p \end{bmatrix}^T \begin{bmatrix} 1 \\ x_1 \\ x_p \end{bmatrix} = \sum_{i=0}^p w_i x_i$$

bias $\leftarrow x_0 = 1$

$$\Rightarrow \hat{y} = w_0 x_0 + w_1 x_1 + w_p x_p$$

$x_0 = 1$

$$\vec{w} = \begin{bmatrix} 383,1010 \\ -3,938 \\ -0,112 \end{bmatrix} \Rightarrow \hat{y} = 383,101 - 3,938 x_1 - 0,112 x_p$$

2-b

(b) Use the model to predict wear when $x_1 = 25$ and $x_2 = 1000$. (5 Marks)

باید برای $x_1 = 25$ و $x_p = 1000$ برتبط مدل برتے آمدہ پیش بینی انجام دہم:

$$\left. \begin{aligned} \hat{y} &= 383,1010 - 3,938 x_1 - 0,112 x_p \\ x_1 &= 25 \\ x_p &= 1000 \end{aligned} \right\}$$

$$\Rightarrow \hat{y} = 383,101 - 3,938 \overbrace{(25)}^{x_1} - 0,112 \overbrace{(1000)}^{x_p} = 180,181$$

2-C

(c) Fit a multivariate linear regression model with an interaction term (x_1x_2) to these data.

موضوعات یک مدل رگرسیون با جمله x_1x_2 طراحی کنید:

ابتدا x_1, x_2 را به ماتریس X اضافه می‌کنیم و ماتریس Z را در دست داریم:

$$Z = \begin{bmatrix} 1 & x_1 & x_2 & x_1x_2 \\ 1 & 1.2 & 1.51 & (1.2)(1.51) \\ 1 & 1.5 & 1.14 & (1.5)(1.14) \\ 1 & 2.2 & 1.51 & (2.2)(1.51) \\ 1 & 2.2 & 1.2 & (2.2)(1.2) \\ 1 & 3.2 & 1.357 & (3.2)(1.357) \\ 1 & 4.0 & 1.115 & (4.0)(1.115) \end{bmatrix} = \begin{bmatrix} 1 & 1.2 & 1.51 & 1.812 \\ 1 & 1.5 & 1.14 & 1.71 \\ 1 & 2.2 & 1.51 & 3.322 \\ 1 & 2.2 & 1.2 & 2.64 \\ 1 & 3.2 & 1.357 & 4.3504 \\ 1 & 4.0 & 1.115 & 4.46 \end{bmatrix}$$

Multivariate Regression

↓
Least Squares →

$$w = (Z^T Z)^{-1} Z^T y$$

$$\hat{y} = w^T z$$

$$Z = \begin{bmatrix} 1 \\ x_1 \\ x_2 \\ x_1x_2 \end{bmatrix}, \quad w = \begin{bmatrix} w_0 \\ w_1 \\ w_2 \\ w_3 \end{bmatrix}$$

① محاسبه $(Z^T Z)^{-1}$:

$$(Z^T Z)^{-1} = \begin{bmatrix} 99.18 & -2.375 & -0.0774 & 0.0025 \\ -2.375 & 0.0106 & 0.0024 & -0.001 \\ -0.0774 & 0.0024 & 0.0009 & -0.00003 \\ 0.0025 & -0.001 & -0.00003 & 1.192 \times 10^{-7} \end{bmatrix}$$

$$B = Z^T y = \begin{bmatrix} 1.4 \\ 2.6171 \\ 1.0100 \\ 22424224 \end{bmatrix}$$

② $z^T y$ محاسبه

$$w = (Z^T Z)^{-1} Z^T y = \begin{bmatrix} 213,944 \\ -1,454 \\ -0,222 \\ 0,004 \end{bmatrix}$$

③ محاسبه بردار w

④ $\hat{y}$ محاسبه

$$\hat{y} = w^T Z = \begin{bmatrix} w_0 \\ w_1 \\ w_2 \\ w_3 \end{bmatrix}^T \begin{bmatrix} 1 \\ x_1 \\ x_2 \\ x_1 x_2 \end{bmatrix} = w_0 + w_1 x_1 + w_2 x_2 + w_3 x_1 x_2$$

$$\Rightarrow \hat{y} = 213,944 - 1,454 x_1 - 0,222 x_2 + 0,004 x_1 x_2$$

Question 3 (30 Marks) ❌❌❌

In a regression problem, we predict O based on one n -dimensional vector (sample) X using the following relation:

$$o = w_0 + w_1x_1 + w_1x_1^3 + w_2x_2 + w_2x_2^3 + \dots + w_nx_n + w_nx_n^3$$

(a) Suppose that we have m samples in our dataset. Write the Matrix form of $O=ZW$ with complete displays of O , Z , and W matrices (Z is constructed based on X) (20 Marks)

❌❌ (b) If we want to obtain W using the gradient-descent method, write the gradient descent relation for this problem such as below (10 marks)

$$w_i \leftarrow w_i + \dots \text{ for } 1 \leq i \leq n.$$

The update relation is not required for the bias term (w_0)

3-a

$$o = w_0 + w_1x_1 + w_1x_1^3 + w_2x_2 + w_2x_2^3 + \dots + w_nx_n + w_nx_n^3$$

ابتدا سعی میکنیم o را prediction کنیم، $\hat{y}_j = o_j = w^T z_j$ بنویسیم

$$o = w_0 + w_1(x_1 + x_1^3) + w_2(x_2 + x_2^3) + \dots + w_n(x_n + x_n^3)$$

$$O = \begin{bmatrix} w_0 \\ w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix}^T \begin{bmatrix} 1 & x_1 + x_1^3 & x_2 + x_2^3 & \dots & x_n + x_n^3 \\ x_1 + x_1^3 & x_2 + x_2^3 & \vdots & & x_n + x_n^3 \end{bmatrix} \Rightarrow \vec{Z} = \begin{bmatrix} 1 & x_1 + x_1^3 & x_2 + x_2^3 & \dots & x_n + x_n^3 \end{bmatrix}$$

$$\Rightarrow \phi_i(x^{(j)}) = (x_{ji} + x_{ji}^3)$$

$$O = ZW$$

(j) x_{ji} یا x_{ji}^3 $\left\{ \begin{array}{l} (0 \leq j \leq m) \text{ Sample اندیس} \\ (1 \leq i \leq n) \text{ Feature شماره} \end{array} \right.$

$$O \in \mathbb{R}^m \quad Z \in \mathbb{R}^{m \times (n+1)} \quad W \in \mathbb{R}^{n+1}$$

$$\begin{bmatrix} o_1 \\ o_p \\ \vdots \\ o_m \end{bmatrix} = \begin{bmatrix} 1 & x_{11} + x_{11}^3 & x_{12} + x_{12}^3 & \dots & x_{1n} + x_{1n}^3 \\ 1 & x_{p1} + x_{p1}^3 & x_{p2} + x_{p2}^3 & \dots & x_{pn} + x_{pn}^3 \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{m1} + x_{m1}^3 & x_{m2} + x_{m2}^3 & \dots & x_{mn} + x_{mn}^3 \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_n \end{bmatrix}$$

m تعداد Sample ها

m اندیس Sample n اندیس Feature شماره

feature شماره = n

3-6

فرض کنید λ نرخ یادگیری باشد Learning Rate.

Gradient Descent: $w_i \leftarrow w_i - \lambda \frac{\partial J(w)}{\partial w_i}$

① $J(w) = \sum_{j=1}^m (y^{(j)} - \hat{y}^{(j)})^2$ ← Cost function

② $\hat{y}^{(j)} = w_0 + \sum_{i=1}^n w_i (x_i^{(j)} + (x_i^{(j)})^3)$

$y^{(j)}$ → خروجی نمونه
 x_i → i امین ورودی (ویژگی)

① جاگذاری $J(w)$

$$\frac{\partial J(w)}{\partial w_i} = \frac{\partial}{\partial w_i} \left(\sum_{j=1}^m (y^{(j)} - \hat{y}^{(j)})^2 \right) = \sum_{j=1}^m \frac{\partial}{\partial w_i} (y^{(j)} - \hat{y}^{(j)})^2$$

$$= 2 \sum_{j=1}^m (y^{(j)} - \hat{y}^{(j)}) \cdot \frac{-\partial \hat{y}^{(j)}}{\partial w_i}$$

جاگذاری $\hat{y}^{(j)}$ از ②

$$= -2 \sum_{j=1}^m \left[(y^{(j)} - \hat{y}^{(j)}) \frac{\partial}{\partial w_i} \left(w_0 + \sum_{k=1}^n w_k (x_k^{(j)} + (x_k^{(j)})^3) \right) \right]$$

چون نسبت به w_i مشتق گرفته‌ایم
برای مقادیر $k \neq i$ به جز $k=i$ صفر می‌شود

$$= -2 \sum_{j=1}^m (y^{(j)} - \hat{y}^{(j)}) (x_i^{(j)} + (x_i^{(j)})^3)$$

$$\Rightarrow \frac{\partial J(w)}{\partial w_i} = -2 \sum_{j=1}^m (y^{(j)} - \hat{y}^{(j)}) (x_i^{(j)} + (x_i^{(j)})^3)$$

Gradient Descent: $w_i \leftarrow w_i - \lambda \frac{\partial J(w)}{\partial w_i}$

$$w_i \leftarrow w_i + 2\lambda \sum_{j=1}^m (y^{(j)} - \hat{y}^{(j)}) (x_i^{(j)} + (x_i^{(j)})^3) \quad \forall 1 \leq i \leq n$$

جالیاری $\hat{y}^{(j)}$ از $\otimes$

$$\omega_i \leftarrow \omega_i + \eta \lambda \sum_{j=1}^m (y^{(j)} - (\omega_0 + \sum_{i=1}^n \omega_i (x_i^{(j)} + (x_i^{(j)})^3))) (x_i^{(j)} + (x_i^{(j)})^3)$$

$\forall 1 \leq i \leq n$